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Research

Mathematics; Algebraic geometry, number theory, representation theory; Shimura varieties, Langlands program, geometric representation theory.

Employment

- 2025– *Professeur des universités*, Université Sorbonne Paris Nord.
- 2024–25 *Akademischer Rat auf Zeit*, Universität Münster.
- 2022–23 *Wissenschaftlicher Mitarbeiter*, Universität Münster.
- 2022 *Visitor*, Max-Planck-Institut für Mathematik, Bonn.
- 2020–21 *Research associate*, Imperial College London.
- 2017–20 *Wissenschaftlicher Mitarbeiter*, Rheinische Friedrich-Wilhelms-Universität Bonn.

Education

- 2020 *PhD in Mathematics*, Universität Bonn.
Advisor: Peter Scholze.
- 2017 *MSc in Mathematics*, Universität Bonn.
Advisor: Peter Scholze.
- 2015 *BSc in Mathematics*, Universidade do Porto.

Papers

- 2025 17. “The Bezrukavnikov equivalence in mixed characteristic”, with K. Bando, I. Gleason, and J. Yu, in preparation.
- 2025 16. “Moduli descriptions of local models”, with T. Richarz, E. Viehmann, and T. Wedhorn, in preparation.
- 2025 15. “On the schematic and analytic constructions of the local Langlands category”, with A. Ivanov, I. Gleason, L. Hamann, and K. Zou, in preparation.
- 2025 14. “Mod p sheaves on Witt flags”, with R. Cass, [arXiv:2503.01796](#).
- 2024 13. “A modular ramified Satake equivalence”, with P. Achar, T. Richarz, and S. Riche, [arXiv:2403.10651](#), submitted.
- 2023 12. “Distributions and normality theorems”, [arXiv:2312.17121](#), submitted.

- 2023 II. “Gaitsgory’s central functor and the Arkhipov–Bezrukavnikov equivalence in mixed characteristic”, with J. Anschütz, Z. Wu, and J. Yu, [arXiv:2311.04043](#), submitted.
- 2022 10. “Fixed points under pinning-preserving automorphisms of reductive group schemes”, with P. Achar, T. Richarz, and S. Riche, [arXiv:2212.10182](#), to appear in *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*.
- 2022 9. “On the connectedness of p-adic period domains”, with I. Gleason, [arXiv:2210.08625](#), submitted.
- 2022 8. “Singularities of local models”, with N. Fakhruddin, T. Haines, and T. Richarz, [arXiv:2208.12072](#), *Mathematische Annalen* **391** (2025), no. 4, 6205–6250.
- 2022 7. “Tubular neighborhoods of local models”, with I. Gleason, [arXiv:2204.05526](#), *Duke Mathematical Journal* **173** (2024), no. 4, 723–743.
- 2022 6. “On the p-adic theory of local models”, with J. Anschütz, I. Gleason, and T. Richarz, [arXiv:2201.01234](#), to appear in *Annals of Mathematics*.
- 2020 5. “On the normality of Schubert varieties: remaining cases in positive characteristic”, with T. Haines and T. Richarz, [arXiv:1806.11001](#), *Annales Scientifiques de l’École Normale Supérieure* (4) **57** (2024), no. 3, 895–959.
- 2020 4. “Théorie de Bruhat–Tits, grassmanniennes affines et modèles locaux”, PhD thesis. *Landesbibliothek Uni Bonn*, [bonndoc](#).
- 2020 3. “Théorie de Bruhat–Tits pour les groupes quasi-réductifs”, [arXiv:2001.05362](#), *Journal de l’Institut de Mathématiques de Jussieu* **21/4** (2022), 1331–1362.
- 2019 2. “Grassmanniennes affines tordues sur les entiers”, [arXiv:1912.11918](#), *Forum of Mathematics Sigma* **11** (2023), Paper no. e12, 65.
- 2017 1. “The Riemannian Hebbbarkeitssätze for pseudo-rigid spaces”, [arXiv:1711.06903](#).

Funding

- 2024–25 Collaborative Research Centre 1442 “Geometry: Deformations and Rigidity”, funded by the Deutsche Forschungsgemeinschaft. Principal investigator in projects A1 (with E. Hellmann and P. Schneider) and A5 (with E. Viehmann and Y. Zhao).

Talks

- 12.12.25 “Comparison of sheaves on Isoc and Bun”, Paris Nord.
- 02.12.25 “Comparison of sheaves on Isoc and Bun”, Orsay.
- 18.11.25 “Comparison of sheaves on Isoc and Bun”, Marseille.
- 10.11.25 “Mod p sheaves on Witt flags”, Singapore.
- 29.09.25 “Comparison of sheaves on Isoc and Bun”, Jussieu.
- 29.08.25 “Comparison of sheaves on Isoc and Bun”, Paderborn.
- 10.04.25 “Comparison of sheaves on Isoc and Bun”, Strasbourg.
- 25.03.25 “Comparison of sheaves on Isoc and Bun”, Clermont-Ferrand.
- 07.10.24 “Arkhipov–Bezrukavnikov for p-adic groups”, Tokyo.
- 22.04.24 “Arkhipov–Bezrukavnikov for p-adic groups”, Jussieu.
- 18.04.24 “Modular ramified Satake”, Münster.

- 23.10.23 “Arkhipov–Bezrukavnikov for p-adic groups”, Geometric and categorical representation theory, Clermont-Ferrand.
- 02.10.23 “Local models revisited”, 75th anniversary of M. Rapoport, Münster.
- 26.07.23 “Teoria geométrica das representações e geometria p-ádica”, Coimbra.
- 26.06.23 “Variétés des drapeaux et +-régularité globale”, Théorie des représentations à Lyon.
- 12.6.23 “Towards Bezrukavnikov via p-adic central sheaves”, Local Langlands and p-adic methods, Bonn.
- 03.02.23 “Towards Bezrukavnikov via p-adic local models”, Oberwolfach.
- 25.10.22 “p-adic local models II”, Münster.
- 18.10.22 “p-adic local models I”, Münster.
- 08.06.22 “Modèles locaux p-adiques: géométrie”, Paris-Nord.
- 01.06.22 “Modèles locaux p-adiques: cohomologie”, Paris-Nord.
- 26.05.22 “Mini-cours sur la théorie p-adique des modèles locaux”, Caen.
- 06.05.22 “Tubular neighborhoods of local models II”, MPIM Bonn.
- 28.04.22 “Local models for p-adic shtukas”, MPIM Bonn.
- 15.03.22 “On the p-adic theory of local models”, Hong Kong.
- 09.02.22 “On the p-adic theory of local models II”, Rampage.
- 12.01.22 “On the p-adic theory of local models”, München.
- 07.12.21 “Sur la théorie p-adique des modèles locaux”, Orsay.
- 29.11.21 “Sur la théorie p-adique des modèles locaux”, Paris Rive Gauche.
- 26.10.21 “On the p-adic theory of local models”, Fields Medal Symposium, Toronto.
- 11.11.20 “Towards the Scholze–Weinstein conjecture on local models”, London.
- 09.10.20 “Vers la conjecture de Scholze–Weinstein sur les modèles locaux”, Paris Nord.
- 24.07.20 “On the geometry of mixed characteristic affine Grassmannians”, Oberwolfach.
- 28.11.19 “Twisted affine Grassmannians in wildly ramified cases”, Bonn.
- 21.11.19 “Bruhat–Tits theory for pseudoreductive groups”, Bonn.
- 21.10.19 “Twisted affine Grassmannians over $\mathbf{Z}$ ”, Modularity and Moduli Spaces, Oaxaca.
- 11.10.19 “Twisted affine Grassmannians and local models of Shimura varieties”, London.
- 30.08.19 “Twisted Kac–Moody groups over the integers”, Luminy.
- 20.12.18 “Local models for some wildly ramified groups”, Bonn.
- 27.04.17 “The Riemannian Hebbbarkeitssatz for pseudo-rigid spaces II”, Bonn.
- 20.04.17 “The Riemannian Hebbbarkeitssatz for pseudo-rigid spaces I”, Bonn.

Conferences

- 08.25 “ENTR Workshop 25”, Paderborn, Germany (speaker).
- 04.25 “Oberwolfach Seminar on Algebraic groups”, Oberwolfach, Germany.
- 03.25 “Simons Collaboration on Perfection in Algebra, Geometry and Topology Annual Meeting” New York, US.
- 03.25 “Geometric approaches to the local Langlands program” Maryland, US.
- 10.24 “Reduction of arithmetic varieties” Oberwolfach, Germany (speaker).
- 10.24 “Workshop on Shimura varieties, representation theory and related topics” Tokyo, Japan (speaker).

- 03.24 “Simons Collaboration on Perfection in Algebra, Geometry and Topology Annual Meeting” New York, US.
- 10.23 “Geometric and categorical representation theory” Clermont-Ferrand, France (speaker).
- 10.23 “Conference on the occasion of Michael Rapoport’s 75th birthday” Münster, Germany (speaker).
- 07.23 “Local Langlands and p-adic methods” Bonn, Germany (speaker).
- 02.23 “Arithmetic of Shimura varieties” , Oberwolfach, Germany (speaker).
- 07.22 “Summer School on The Langlands Programme” , Paris, France.
- 05.22 “30e Rencontres arithmétiques de Caen” , Caen, France (speaker).
- 11.21 “2021 Fields Medal Symposium: Peter Scholze” , Toronto, Canada (speaker).
- 07.21 “Oberwolfach Arithmetic Geometry” , Oberwolfach, Germany (speaker).
- 10.19 “Modularity and moduli spaces” , Oaxaca, Mexico (speaker).
- 08.19 “Hausdorff School on the Emerton-Gee stack and related topics” , Bonn, Germany.
- 09.19 “Immeubles et grassmanniennes affines” , Luminy, France (speaker).
- 06.19 “Arithmetic Geometry in Carthage” , Carthage, Tunisia.
- 05.19 “The p-adic Langlands programme and related topics” , London, England.
- 06.18 “Groupes algébriques et géométrisation du programme de Langlands” , Lyon, France.
- 04.17 “Leçons Hadamard par P. Scholze” , Paris, France.
- 03.17 “Arizona Winter School 2017: Perfectoid Spaces” , Tucson, USA.
- 10.16 “Oberwolfach Seminar on Perfectoid Spaces” , Oberwolfach, Germany.

Teaching

- 2026 Course “Espaces perfectoïdes” , Paris Diderot.
- 2026 Tutorial “Résumé de cours d’algèbre linéaire” , with C. Ausoni, Paris Nord.
- 2025 Tutorial “Travaux dirigés d’algèbre commutative” , with P. Boyer, Paris Nord.
- summer 25 Working group “Affine Deligne–Lusztig theory” , with E. Viehmann and I. Zachos.
- summer 25 Course “Lineare Algebra II” , teaching assistant of E. Viehmann.
- winter 24 Working group “Vector bundles in the v-topology and Sen theory” , with E. Hellmann and L. Mann.
- winter 24 Course “Lineare Algebra I” , teaching assistant of E. Viehmann.
- summer 24 Course “Algebraic Geometry II” .
- winter 23 Course “Geometrische Lineare Algebra” , teaching assistant of U. Hartl.
- summer 23 Seminar “Homological and Frobenius methods in commutative algebra” .
- winter 22 Working group “Bezrukavnikov” , with K. Zou.
- winter 22 Seminar “Darstellungstheorie endlicher Gruppen” , with E. Viehmann.
- winter 20 Working group “Geometrization of the Langlands program” , with M. Tamiozzo.

Visits

- 10.25 Singapore , with I. Gleason.
- 03.25 Maryland with I. Gleason and J. Yu, hosted by T. Haines.
- 07.22 Grenoble, with P. Achar and S. Riche.

06.22 Paris-Nord, with S. Morra and A.-C. Le Bras.
Fall 21 Paris-Saclay, with K. Česnavičius.
02.20 Imperial College London, with A. Caraiani.
12.18 Jussieu, with J. Anschütz and T. Richarz.

Referee

Annals of Mathematics, Inventiones Mathematicae, Forum of Mathematics Pi, Annales scientifiques de l'École Normale Supérieure, Mathematische Annalen, Representation theory, Canadian Journal of Mathematics, Épijournal de Géométrie Algébrique.

Languages

Portuguese (native), German, English, French, Spanish (fluent), Italian (advanced).

References

Thomas Haines, Timo Richarz, Simon Riche, Peter Scholze, Eva Viehmann.